

PUBLIC VS PRIVATE FIRM SIZE IN SINGAPORE

A FIVE-YEAR STATISTICAL COMPARISON OF REVENUE AND TOTAL ASSETS

DS607 - GROUP PRESENTATION

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Research Questions

1

Do the revenue and total asset distributions of public and private Singaporean companies differ over a 5 year period (2021-2025)

2

what are the annual median differences and their 95% confidence intervals?

Data Source & Series

Data Source

ORBIS

Time Period

2021-2025

Data Series

Operating Revenue
Total Assets

DATA CLEANING & SAMPLE CONSTRUCTION

- Listing status defines the groups: Listed = public, Unlisted = private; delisted firms dropped
- Consolidation codes are filtered to ensure comparable financial reporting levels.
- Negative revenue and asset values were excluded for consistency with accounting comparability
- Missing values are excluded separately by measure and year; sample sizes vary with data availability.

479

Public Firms

33,228

Private Firms

STATISTICAL METHODS

Descriptive statistics

Sample size, median, Q1, Q3 and IQR by year, measure and group; box plots on a log scale

ECDF plots

Empirical cumulative distributions of revenue and total assets, public vs private, one panel per year

Two-sample KS test

H0: same distribution in that year. H1: distributions differ. Statistic D = largest gap between the two ECDFs; 10 tests (2 measures × 5 years)

Bootstrap median difference

10,000 resamples per group; 95% percentile CI for public minus private median.

Descriptive Statistics

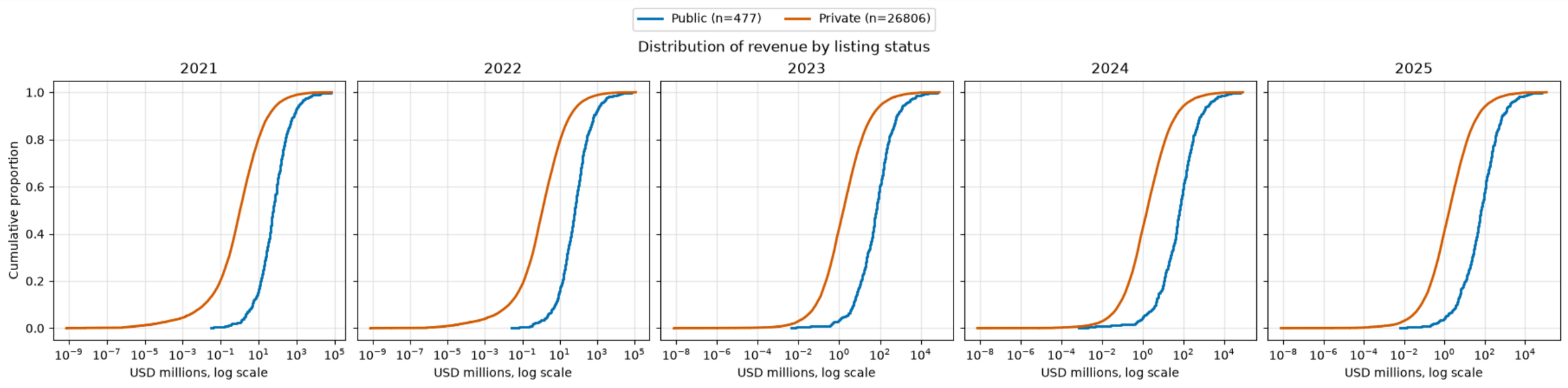
Year	Measure	Group	Sample size	Median	Q1	Q3	IQR
2021	Revenue	Public	477	57.32	17.83	185.10	167.27
2021	Revenue	Private	26,806	1.03	0.16	6.14	5.97
2021	Total assets	Public	479	111.52	38.66	589.36	550.70
2021	Total assets	Private	33,015	2.34	0.31	17.90	17.59
2022	Revenue	Public	478	64.92	18.89	190.66	171.78
2022	Revenue	Private	26,989	1.20	0.20	7.12	6.92
2022	Total assets	Public	479	110.01	35.90	591.93	556.03
2022	Total assets	Private	33,042	2.62	0.36	19.17	18.81
2023	Revenue	Public	478	64.33	17.60	208.61	191.01
2023	Revenue	Private	23,245	1.61	0.33	8.15	7.82
2023	Total assets	Public	479	104.52	35.00	621.78	586.77
2023	Total assets	Private	33,010	2.73	0.38	19.72	19.34
2024	Revenue	Public	479	63.13	16.13	216.31	200.18
2024	Revenue	Private	23,122	1.63	0.33	8.52	8.19
2024	Total assets	Public	479	105.73	34.43	644.52	610.08
2024	Total assets	Private	32,987	2.66	0.36	19.49	19.13
2025	Revenue	Public	479	63.29	16.98	234.33	217.36
2025	Revenue	Private	22,719	1.75	0.35	9.26	8.91
2025	Total assets	Public	479	118.98	34.52	715.09	680.57
2025	Total assets	Private	32,902	2.69	0.36	19.65	19.29

USD mns; positive values only

- Revenue: Public median remains around USD 57-65m; private median rises from USD 1.03m to USD 1.75m.
- Total assets: Public median ranges from USD 105-119m; private median remains around USD 2-3m.
- Public firms' median revenue and assets are approximately 35-60× larger.

MAIN FIGURES · ECDF

Revenue: public firms sit to the right

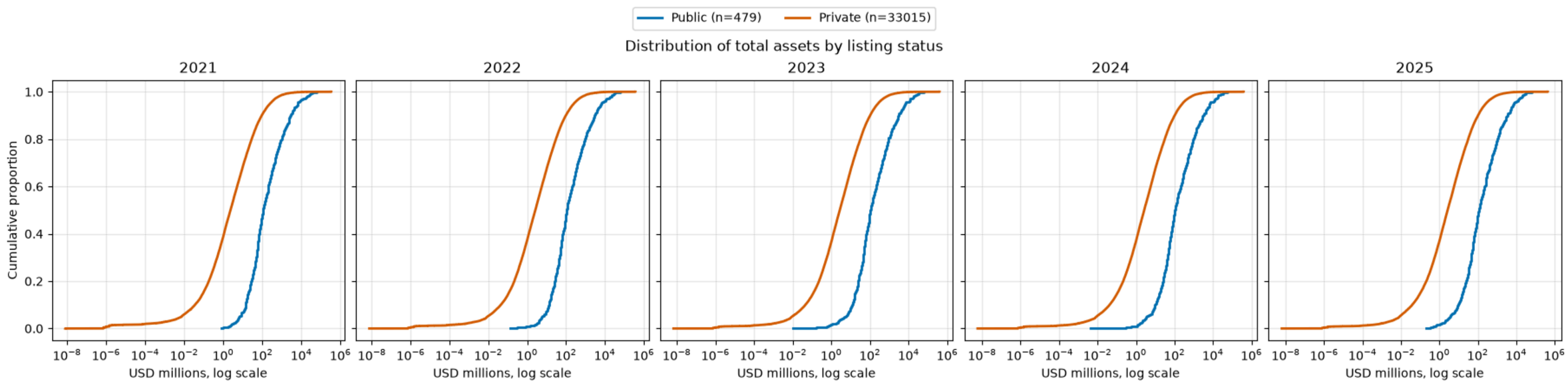


Public firms are larger in every year. The blue curve sits right of the orange across almost the whole range.

The gap is widest in the middle. Curves nearly meet at the very top and bottom, so the difference is in the typical firm.

MAIN FIGURES · ECDF

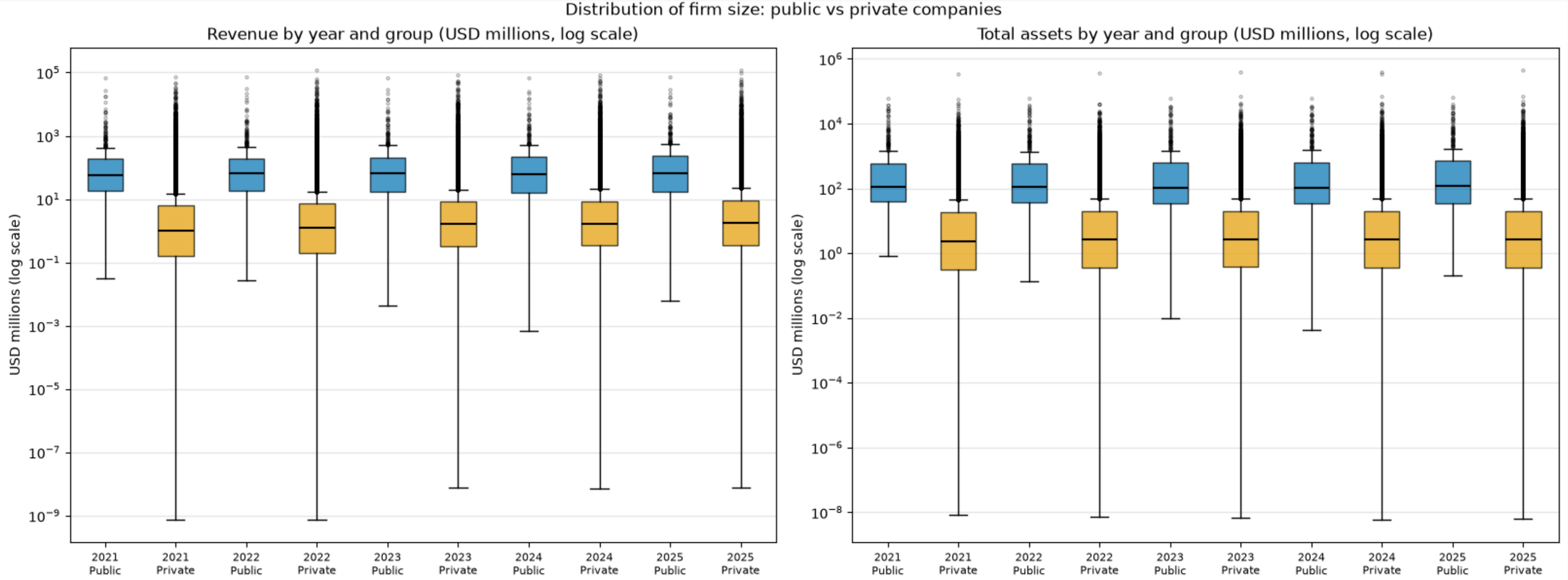
Total assets: same pattern



Public asset distributions are more concentrated, with steeper ECDF curves.

Private firms exhibit greater dispersion and a long lower tail of extremely small asset values.

Box plots: the public-private gap is large and persistent



- Median positions remain broadly stable across 2021-2025, indicating a persistent size gap.
- Private firms exhibit greater size dispersion and more extreme observations.

STATISTICAL METHODS · KS TEST

What does the KS Test answer?

REVENUE:

Null Hypothesis (H_0): Public and private firms have the exact same revenue distribution for that year.

Alternative Hypothesis (H_1): The revenue distributions for public and private firms are different for that year.

TOTAL ASSETS:

Null Hypothesis (H_0): Public and private firms have the exact same distribution of total assets for that year.

Alternative Hypothesis (H_1): The distributions of total assets for public and private firms are different for that year.

Step 1

ECDFs

For each group, the share of firms at or below every value.

Step 2

Find D

D = the largest vertical gap between the two ECDFs.

Step 3

Test

scipy ks_2samp, two-sided, gives D and its p-value.

Repeated 10 times: 2 measures × 5 years (2021-2025).

Decision: small p-value → reject H_0 that the two distributions are the same.

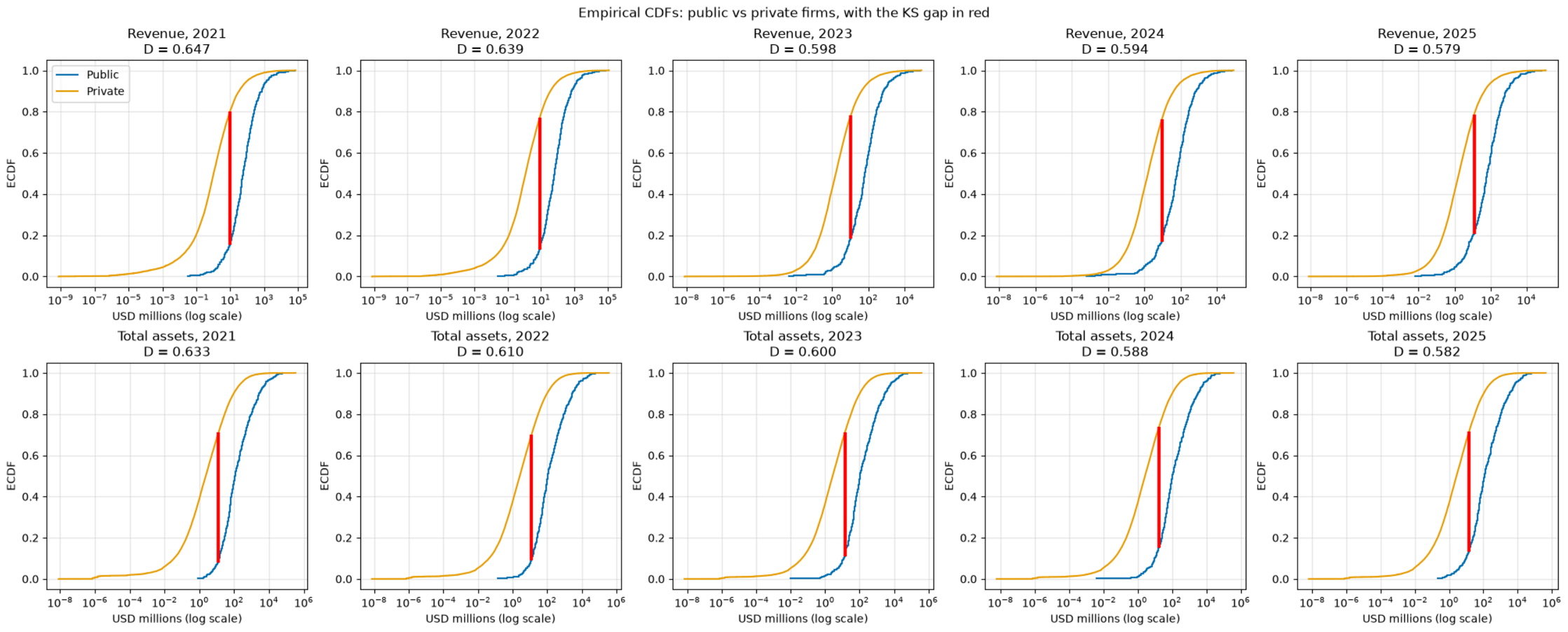
MAIN FIGURES · KS TEST

KS test: all 10 comparisons reject equal distributions

Year	Revenue D	Assets D	p-value
2021	0.647	0.633	< 0.001
2022	0.639	0.610	< 0.001
2023	0.598	0.600	< 0.001
2024	0.594	0.588	< 0.001
2025	0.579	0.582	< 0.001

- D falls from 0.647 to 0.579 for revenue and 0.633 to 0.582 for total assets
- With about 479 public vs 23,000 to 33,000 private firms, the test detects even minor differences

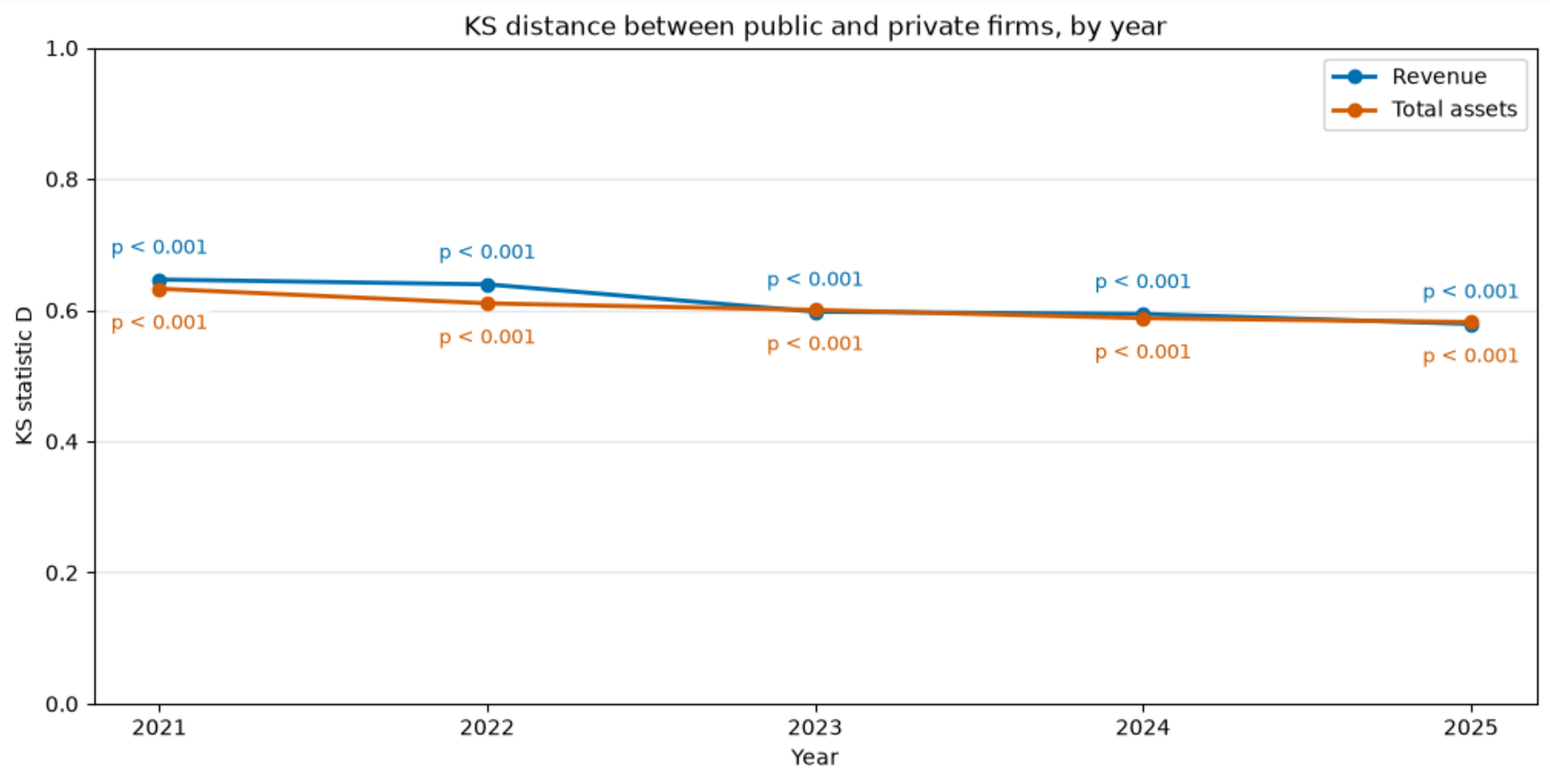
The largest gap is at typical firm size, USD 10-20m



Gap is almost always near USD 10 to 20 mn: 70-80% of private firms are smaller there, vs 8-21% of public firms

The two groups differ by roughly 60pp at their most separated point

KS distance narrows slowly, but never closes



- D starts near 0.65 in 2021 and falls to about 0.58 by 2025
- The fall of about 0.05 to 0.07 is small next to the overall separation: converging slowly, not coming together
- Every test is significant at the 0.001 level in all years

Why Bootstrap

We used bootstrap resampling to estimate uncertainty around a median difference without assuming that the data follows a normal distribution.

STATISTICAL METHODS · BOOTSTRAP CONFIDENCE INTERVALS

How the Bootstrap Confidence interval were constructed

Step 1

Filter

Same samples as the KS test: positive, non-missing values per year and measure.

Step 2

Estimate

Point estimate = public median minus private median.

Step 3

Resample

Draw each group with replacement, at its original size.

Step 4

Repeat

10,000 times (seed 2026), recording the median difference each time.

Step 5

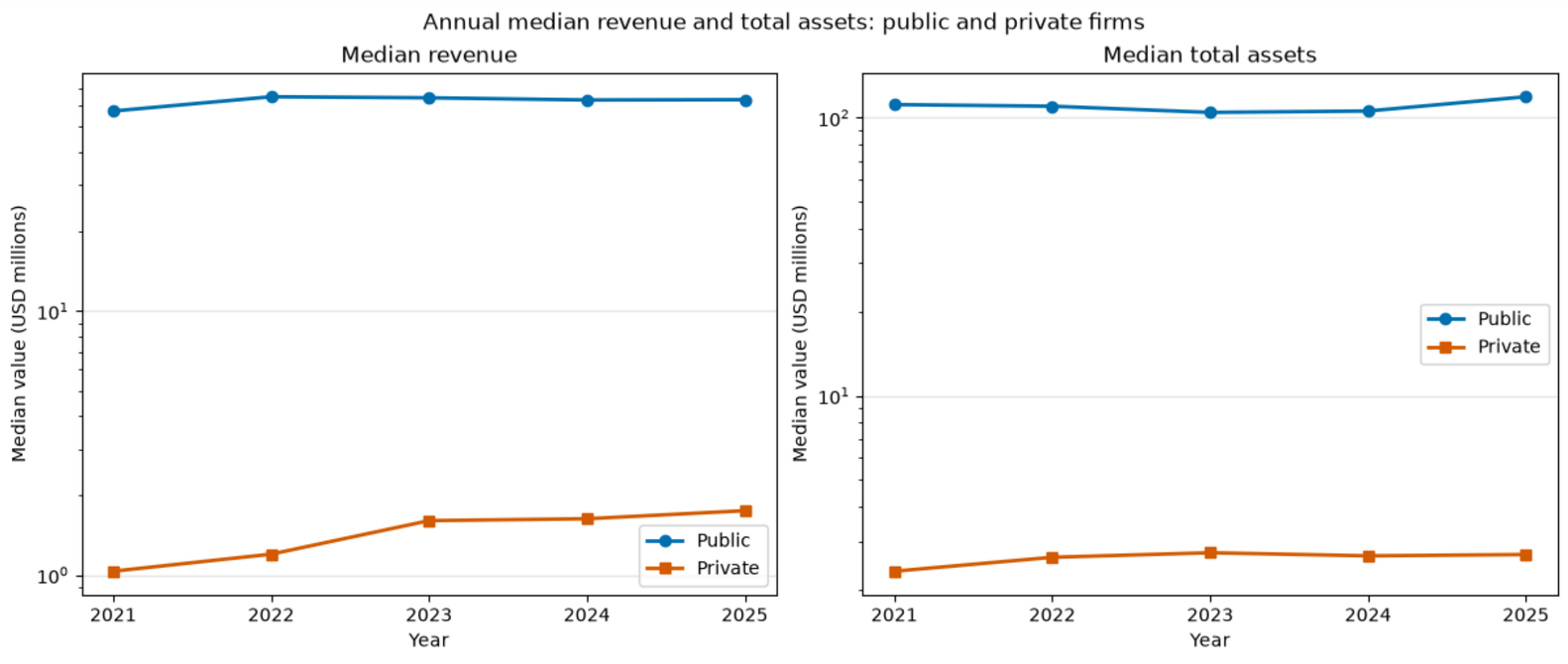
95% CI

2.5th and 97.5th percentiles of the 10,000 differences.

Repeated 10 times: 2 measures × 5 years (2021-2025).

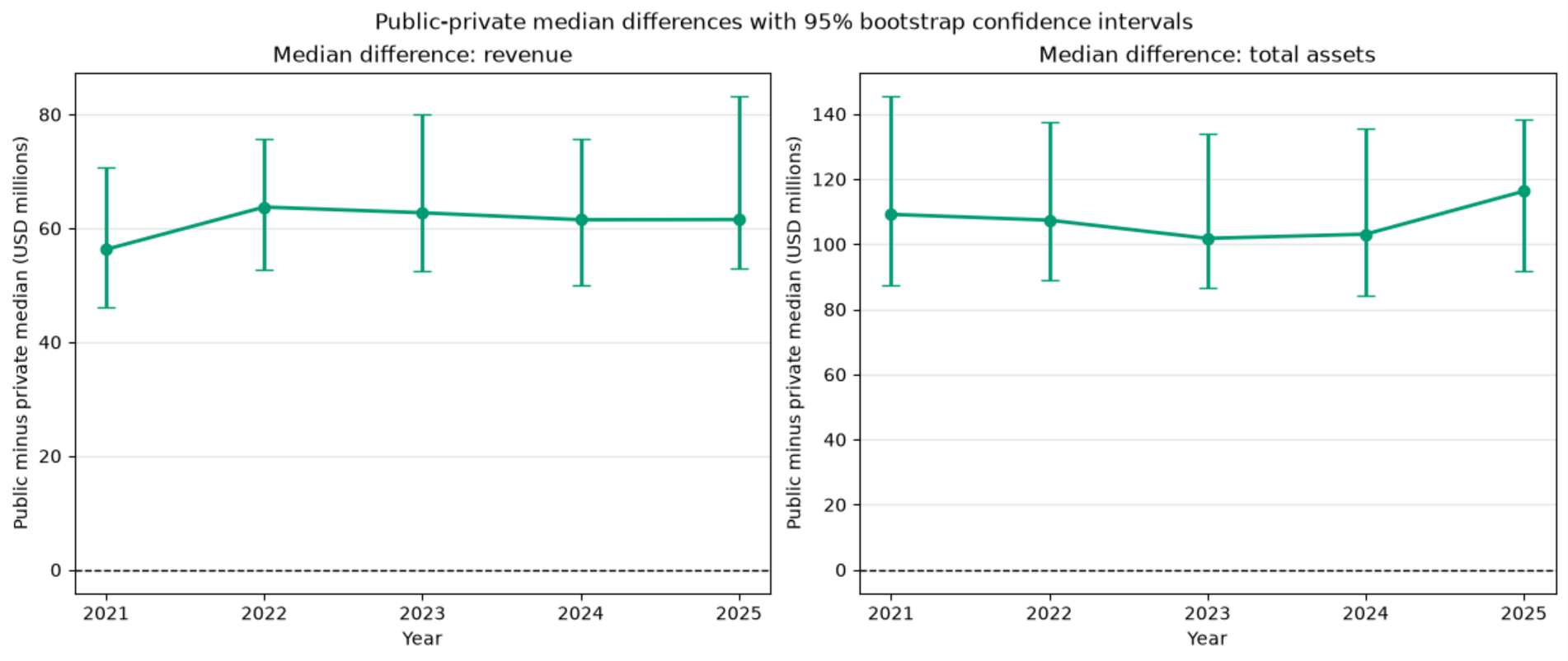
interpretation: if the 95% confidence interval excludes 0, the data support a non-zero public-private median difference

Public firms have substantially higher median revenue and total assets



- Revenue: public median about USD 57-65m vs private about USD 1-2m
- Total assets: public about USD 105-120m vs private about USD 2.5m
- Public medians are broadly stable; private revenue median almost doubles, private assets stay flat

Every confidence interval is above zero



- Every 95% confidence interval is above zero
- Public median is larger in every year, for both measures
- Gap is about USD 60m for revenue and about USD 100m for total assets, broadly stable across years

MAIN FIGURES · BOOTSTRAP TABLE

Median public-private gap & 95% bootstrap CI, USD millions

Measure	Year	Public median	Private median	Difference	95% CI
Revenue	2021	57.32	1.03	56.29	[46.03, 70.68]
Revenue	2022	64.92	1.20	63.73	[52.80, 75.62]
Revenue	2023	64.33	1.61	62.72	[52.38, 80.10]
Revenue	2024	63.13	1.63	61.50	[49.90, 75.65]
Revenue	2025	63.29	1.75	61.54	[52.87, 83.15]
Total assets	2021	111.52	2.34	109.18	[87.41, 145.38]
Total assets	2022	110.01	2.62	107.39	[89.18, 137.40]
Total assets	2023	104.52	2.73	101.79	[86.69, 133.84]
Total assets	2024	105.73	2.66	103.08	[84.16, 135.74]
Total assets	2025	118.98	2.69	116.29	[91.62, 138.13]

Conclusion

Magnitude of the gap

Public firms' median revenue and assets exceed private firms substantially

Persistent Distributional Separation

KS statistics confirm substantial separation despite modest five-year convergence

Private-Firm Scale

Private revenue nearly doubles, yet remains far below public medians

Distributional Concentration

Most private firms fall below USD 10-20mn; few public firms do

Limitation

Relative Size Ratios

Large ratios may partly reflect unusually small private-firm medians

Distributional Testing

KS significance does not quantify economic importance or causality

Sample Composition

Changing annual samples may influence observed convergence patterns

Inference Over Time

Annual confidence intervals do not test changes between years

Generalisability

Positive-value filtering excludes distressed and inactive firms

THANK YOU !!!